Chapter 4: Nominal and Effective Interest Rates

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Instructions:

- For every calculation, clearly state the values of the nominal rate r, the compounding frequency m, and the relevant time period before solving.
- Show the formula you are using before substituting numerical values.
- Round final dollar answers to the nearest cent and rate answers to two decimal places, unless otherwise stated.

Exercise 1: Nominal Interest Rate on a Student Savings Account

Karim just opened a savings account at his university's partner bank to set aside money during his studies. The account statement he received does not mention an annual rate anywhere; instead, it simply states that interest accrues at 0.75% every month on the balance.

1. Determine the nominal annual interest rate (APR) implied by this monthly rate.
2. Karim's roommate has a similar account, but his bank statement expresses interest on a per-quarter basis instead. Determine what the nominal rate per quarter would be for Karim's account.

Exercise 2: Nominal to Effective Annual Rate on a Credit Card

A regional bank is launching a new credit card aimed at engineering students. The marketing brochure advertises "only 15% interest per year," but in the fine print, it states that interest is calculated and applied to the outstanding balance at the end of every month.

1. Identify the nominal annual rate r and the compounding frequency m implied by the fine print.
2. Determine the effective annual interest rate (APY) that a cardholder who carries a balance for a full year would actually pay.
3. If a competing card instead advertises "15.5% effective annual rate, compounded annually," determine which of the two cards is actually more expensive for the cardholder.

Exercise 3: Comparing Effective Rates Across Compounding Periods

A certificate of deposit (CD) is a savings product in which a customer locks in a fixed sum for a set term in exchange for a fixed, usually higher, interest rate than a regular savings account. An investment advisor is presenting three CDs to a client, all of which are quoted using the same nominal annual rate of 8%, but each compounds interest differently: CD 1 compounds semi-annually, CD 2 compounds quarterly, and CD 3 compounds monthly.

1. Determine the effective interest rate per compounding period for each CD.
2. Determine the effective annual interest rate for each CD.
3. Based on your results, state and briefly justify the general relationship between compounding frequency and the effective annual rate, assuming the nominal rate stays fixed.

Exercise 4: Continuous Compounding on a Long-Term Reserve Fund

A hospital system maintains an equipment-replacement reserve fund, currently holding \$50,000. Because the fund is actively managed and reinvested on an ongoing basis, the hospital's finance office models the fund as growing under continuous compounding at a nominal annual rate of 6%.

1. Determine the effective annual interest rate implied by this continuous compounding assumption.
2. Determine the future value of the \$50,000 reserve after 8 years under this assumption.
3. A competing fund instead offers monthly compounding at a slightly higher nominal annual rate of 6.15%. Determine the future value of the \$50,000 under this alternative after 8 years, and state which of the two funds would leave the hospital with more money at the end of the 8 years.

Exercise 5: Evaluating Two Equipment Financing Offers

An engineering firm needs to finance a new \$180,000 CNC machining center for its manufacturing line. Two banks submit financing proposals: Bank A structures its offer as "8.4% annual interest, compounded monthly," matching the structure the firm has seen on its other equipment loans. Bank B instead advertises "8.7% per year," stating explicitly that this figure is already an effective annual rate. A third informal option, offered by the equipment manufacturer's in-house financing arm, charges "2.1% every 3 months," with no further explanation of how this accumulates over a year.

1. Determine the effective annual interest rate implied by Bank A's offer.
2. Determine the effective annual interest rate implied by the manufacturer's in-house financing arm.
3. Rank all three financing options from cheapest to most expensive for the firm, based purely on their effective annual interest cost, and justify the ranking numerically.

Exercise 6: Present Worth of a Future Car Purchase

A car dealership is holding a specific vehicle for a customer, Ali, who plans to buy it in 3 years once he finishes his studies. The dealership guarantees today's price will still apply, but Ali must pay the agreed amount in a single payment when he picks up the car in 3 years. To make sure he has enough money ready, Ali wants to set aside a single amount today, in an account that earns an annual interest rate of 10%, compounded monthly, so that the account grows to exactly cover the $60,000 price by the time he needs to pay.

1. Identify the values of r, m, and the total number of compounding periods k that apply to this 3-year growth period.
2. Determine the amount Ali must deposit today so that the account reaches $60,000 in 3 years.
3. If Ali can currently only set aside $45,000 today, determine whether this amount, growing at the same rate, will be enough to cover the $60,000 price, and state the shortfall or surplus.

Exercise 7: Pricing a Corporate Bond Given a Required Yield

A mid-sized medical device manufacturer issued a corporate bond several years ago to fund the construction of a new production facility. The bond has a face value of $5,000, carries an annual coupon rate of 7%, and pays its coupons semi-annually. The bond now has exactly 8 years remaining until its maturity date. An investment analyst evaluating the bond for a client's portfolio determines that, given current market conditions and the manufacturer's credit profile, investors would require a yield to maturity of 9% (a nominal annual rate, compounded semi-annually) in order to be willing to purchase this bond today.

1. Identify the values of F, C, λ, m, and n that apply to this bond, based on the scenario.
2. Draw the cashflow diagram of this bond.
3. Determine the fair market price of the bond today.
4. State, without recalculating, whether this bond is trading at a premium, at a discount, or at par relative to its face value, and briefly justify your answer using the relationship between the coupon rate and the yield to maturity.

Exercise 8: Premium, Discount, and Par Pricing Across Two Bonds

An investment fund is comparing two corporate bonds, each with a face value of $1,000. Bond A was issued with an annual coupon rate of 6%, while Bond B was issued with an annual coupon rate of 9%. Both bonds pay coupons semi-annually and have exactly 10 years remaining until maturity. Given current market conditions, both bonds are being evaluated by investors who require a yield to maturity of 7.5% (nominal annual, compounded semi-annually).

1. Without performing a full price calculation, state whether Bond A and Bond B will each trade at a premium, at a discount, or at par, and justify your answer using the relationship between coupon rate and yield.
2. Calculate the exact price of Bond A today.

3. Calculate the exact price of Bond B today.

4. Suppose that market conditions shift and the required yield to maturity for both bonds rises to 10%. Without recalculating exact prices, explain what will happen to the prices of Bond A and Bond B, and briefly explain why bond prices react this way to a change in required yield.

Exercise 9: Investment Growth Under a Time-Varying, Multi-Structure Rate Agreement

An investor places \$15,000 into a structured investment fund at time 0. Rather than offering a single fixed rate, the fund's prospectus specifies that the applicable interest structure changes every year, as follows: during the first year, the fund pays a nominal annual rate of 8%, compounded quarterly; during the second year, the rate changes to a nominal annual rate of 10%, compounded monthly, reflecting a shift in the fund's underlying investment strategy; and during the third year, the fund switches to a simpler structure and pays a flat effective annual rate of 9%, with no further compounding adjustments needed.

1. Determine the effective annual interest rate that applied during Year 1.

2. Determine the effective annual interest rate that applied during Year 2.

3. Determine the future worth of the investor's \$15,000 at the end of Year 3, taking into account all three differently structured years.

4. Suppose the fund instead offered a single constant annual rate for all 3 years, chosen so that the investor's final future worth after 3 years exactly matches your answer from part (c). Set up (but do not necessarily solve algebraically beyond stating the equation) the relationship that this constant equivalent rate would need to satisfy.